REQUISITES:

- Use real information for the title of your project.

- WRITE IN PASSIVE. Think like if it was or has been done/made; or like it is or will be done/made. Avoid using "I" and "you." Try to avoid the repetitive use of "we."

- Tenses in a sentence may be mixed as long as they make sense, i.e. simple present and simple past, or present perfect and past perfect, or present progressive and past progressive, etc.

- Include grammar and key words from the guide book “English for Science and Engineering.”

NOTE: The verb “make” is used more in actions on physical things or the creation of physical things; the verb “do” is more for the thinking and/or action themselves.

PROJECT FORMAT – only for this course

Title: An explanatory sentence of your project idea, which later may be transformed into a creative short name.

Introduction. A brief history of your title, i.e. why its beginnings, its current state, and end it with the intention of a different implementation to an actual problem. Length = 1/2 page.

General Objective. Why and Where the idea of the innovation or product. 3 lines in Simple Present.
Specific Objectives. Three QUANTITATIVE variables. Use the methodology section of a scientific article.
1 line each in Simple Present.

Unit1.
Research and Development (R&D):
a) Create a timetable (this is not the CPM diagram); b) acquisition of the raw materials: 3 lines; c) professional group of work: 3 lines; d) financial aspect: 4 lines each (search for market vs. time, or sales vs. time); e) a relevant legal norm: 3 lines minimum.

Unit 2.
Design and Testing.
Find a detailed drawing, picture, or image of the object or procedure you chose (if you can use Autocad, Solidedge, Fushion, or other, it helps) to write and talk about the problem and the solution.

Value Engineering. One paragraph of 3 lines about the Function vs. Cost concept - this means writing about the possible benefit(s) of your solution for a company or the society (the four steps of track 11 in the guide book explain the concept).

Testing. This can be confused with experimentation or technical knowledge. To make it easy, just write a possible outcome for each specific objective. The experiments and data in the methodology section of a good scientific article may help you find short descriptions.

For example. in Unit2, lesson 6, point f), a quantitative variable for a microwave oven may be the amount of energy used to thaw a particular kind of food in units of radiation or the watts used in one hour of operation.

Unit 3.

Manufacturing and Industry

In this section you have to choose a sketch or image of the manufacturing plant (also called factory) of the industry that creates and implements parts or procedures related to your career.
Write 4 lines about the roles of the workers inside the factory that follow steps within the buildings with the machines involved in the assembly lines.

Reminder**:**
1. Do not scan something drawn by hand on paper, it is unprofessional.
2. Remember the testing involves beta, semi-open, or open.

Unit 4.
Safety, Maintenance, and Quality Control.

Personal Protective Equipment (PPE). Use an image about the equipment requirements that the workers need to have to work in the manufacturing plant like special clothes, goggles, earplugs, helmets, etc. Write 3 lines describing the safety measures.

Technical difficulty. Write 3 lines about the possible cause of the technical difficulty, and another 2 lines for its consequences.
Was it due to mechanical failure, electrical failure, materials failure, human error, etc.? (rely on real info of an accident in a company.)

Hazard analysis of defects (or low quality). Write 3 lines about some maintenance work to do to the technical difficulty in the manufacturing plant.
Create a PDCA cycle as described in the guide book on page 105, track 28.

Troubleshooting. Write 3 lines about the appropriate engineer post to tackle the technical difficulty. Use track 25 on page 104 which describes the posts of engineers in a manufacturing plant, i.e. operations engineer - maintenance engineer - control engineer - safety engineer.

Unit 5

Careers and employment. Write 4 lines about the professional scope of your career, job payments, and the future specific niche of your project's usefulness. Find a few images of companies where you would like to work.

Innovation and business events. Write 4 lines about the potential plenaries, social events, scientific fairies, or engineering firms in which your product may be presented to do business. Find a few logos or brands of the ones you would like to negotiate with.